

## Project Design Phase-II

### Data Flow Diagram & User Stories

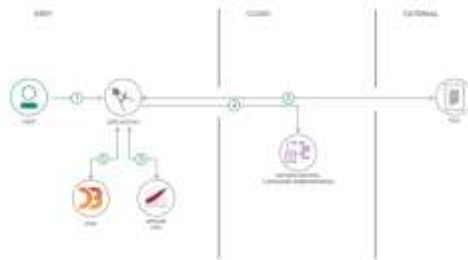
|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| Date          | 25 June 2026                                                      |
| Team ID       | SWTID-2026-4616                                                   |
| Project Name  | Visualization Tool for Electric Vehicle Charge and Range Analysis |
| Maximum Marks | 4 Marks                                                           |

#### Data Flow Diagrams:

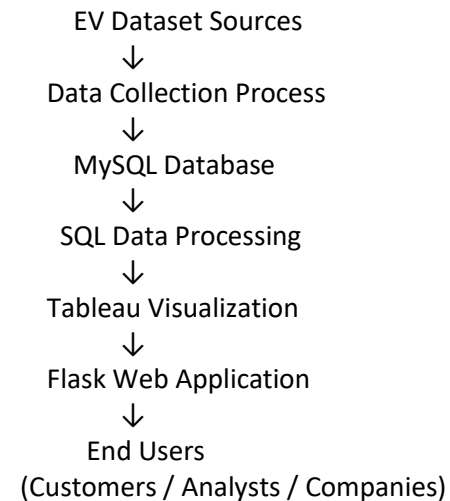
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

#### Example: [\(Simplified\)](#)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.



## User Stories

| User Type     | Functional Requirement  | User Story Number | User Story / Task                                                                          | Acceptance Criteria                           | Priority | Release  |
|---------------|-------------------------|-------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------|----------|----------|
| EV Customer   | Dashboard Visualization | USN-1             | As a user, I can view EV analytics dashboard to understand vehicle insights                | User can access interactive Tableau dashboard | High     | Sprint-1 |
| Data Analyst  | Data Analysis           | USN-2             | As an analyst, I can analyze EV datasets using SQL queries                                 | Insights can be generated from database       | High     | Sprint-1 |
| EV Researcher | EV Comparison           | USN-3             | As a researcher, I can compare EV brands based on price, range, efficiency and performance | Comparison charts are displayed successfully  | High     | Sprint-1 |
| Web User      | Web Integration         | USN-4             | As a user, I can access Tableau dashboards through Flask web interface                     | Dashboard loads successfully in webpage       | Medium   | Sprint-2 |
| Administrator | Data Management         | USN-5             | As an admin, I can update EV datasets for future analysis                                  | New data can be added and processed           | Medium   | Sprint-2 |